Personal Notes on Partition Theory

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Proposition 1. The number of partitions of n in which parts $\lambda_1, \lambda_2, \ldots, \lambda_m$ appear at least once equals the number of partitions of $n - \sum_{i=1}^{m} \lambda_i$.

Proposition 2. The number of partitions of n in which multiples of x must be distinct equals the number of partitions of n in which no part appears more than $2x - 1$ times.

Proposition 3. The number of partitions of n in which multiples of x must be distinct equals the number of partitions of n in which no part is divisible by $2x$.